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SUMMARY

Hydrogen is the simplest atom which contains one electron and one proton. It has three isotopes namely protium, deuterium and tritium. The hydrogen exists as diatomic gaseous molecule. The molecule has two nuclear spin isomers namely ortho and para-hydrogen. Elemental hydrogen having an electronic configuration of $1s^1$, has properties similar to alkali metals as well as halogens. Based on its electronic configuration and existence of the +1 oxidation state in most of the compounds it is placed in group 1 along with alkali metals.

Hydrogen is prepared by the action of metal with mineral acids in laboratory. In industrial scale, hydrogen is produced by the steam-reforming of hydrocarbons. Deuterium can be synthesized by the electrolysis of heavy water, while the radioactive tritium is prepared by the nuclear transmutation reaction of lithium with slow neutrons.

Hydrogen reacts with nonmetals like oxygen and halogens to form water and hydrogen halides respectively. It reacts with metals to form metal hydrides. It acts as good reducing agent. Deuterium undergoes exchange reactions with hydrogen. Tritium with a half-life period of 12.4 years and emits β-particles spontaneously. Hydrogen has many uses including in the preparation of ammonia, methanol, and hydrogenation of oils and reduction of metal oxides. It can also be used as fuel.

Water is an important amphoteric oxide of hydrogen and used a universal solvent. It hydrolyses oxides and forms metal hydroxides with metals such as sodium and barium. Due to the presence of soluble salts of calcium and magnesium in water, the water shows hardness. Clark's method can be used to remove temporary hardness. Like water, another important oxide is hydrogen peroxide. It is used as a mild antiseptic and as a bleaching agent in textile and paper industries. Hydrogen bound to strong electronegative atoms such as fluorine, nitrogen or oxygen atoms, forms a peculiar type of electrostatic interactions with another strong electronegative atom. This is called hydrogen bond. This is one of important interactions observed in many biological molecules such as proteins, DNA etc…